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Co-LLM: A retrieval-augmented language model framework for chiller optimization control

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ABSTRACT

In HVAC systems of large buildings, particularly in regions with significant cooling demands due to hot climates, chillers account for the bulk of energy consumption. Their operation is influenced by a range of factors, including both indoor conditions and outdoor environmental factors. Consequently, optimizing chiller control policies remains a central challenge in building energy conservation. Consequently, optimizing chiller control policies remains a central challenge in building energy conservation, especially in environments where there is a high demand for cooling. To date, few study has combined large language models (LLMs) with domain knowledge for chiller control. This paper proposes a Retrieval-Augmented Cooperative LLM-based Control Framework (Co-LLM). The framework constructs a chiller-specific knowledge base from industry standards, expert experiences, and operational logs. High-quality prior policies are retrieved via a Retrieval-Augmented Generation (RAG) mechanism, after which a Selective Recalibration module locally adjusts only those sub-policies that are strongly affected by external disturbances. Finally, a RAG-tuned Exploration performs multi-round optimization using a composite energy-comfort metric. A field study on the central HVAC system of a tertiary hospital in Taizhou shows that Co-LLM yields a 1.9% energy-saving gain over the traditional expert policy and achieves higher system COP than all other baselines, outperforming mainstream control methods. In particular, the retrieval-augmented mechanism effectively suppresses 92.7% comfort plunges and significantly enhances the robustness of the control system.

1. Introduction

Building energy consumption accounts for roughly 20%–40% of global energy use, with nearly half attributed to heating, ventilation, and air conditioning (HVAC) systems. In large buildings in tropical and subtropical regions, people's demand for cooling often far exceeds their demand for heating. In HVAC systems, the chiller is the main energy-consuming component. Compared to chilled water pumps, fans, and cooling towers, the chiller accounts for more than 50% of HVAC energy consumption [1]. optimizing their operation becomes crucial for achieving substantial energy savings. While passive cooling methods, such as fans, can achieve the same cooling effect as air conditioning in office buildings, thereby reducing energy consumption. [2]. Therefore, improving

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the operational efficiency of chillers is of great significance for energy conservation. However, as HVAC system equipment ages, operational efficiency declines. Upgrading the equipment is theoretically the optimal solution, but it also faces the issue of high costs. Although passive cooling methods, such as fans, can effectively substitute for the cooling effect of chillers [2], optimizing policies in large buildings is widely considered a more economically feasible and practically operable alternative.

In practice, control policy optimization plays a central role in determining how efficiently chillers operate under varying load and environmental conditions. In the domain of HVAC control policy optimization, three representative methods are frequently adopted: Rule-Based Control (RBC), Model-Based Control (MBC), and Reinforcement Learning (RL). Among them, RBC remains the most widely deployed strategy in existing buildings due to its simplicity, low implementation cost, and compatibility with aging HVAC infrastructures. RBC relies on manually designed logical rules, such as predefined thresholds and switching conditions, to regulate chiller operation. However, RBC does not explicitly model the dynamic behavior of HVAC systems, making it challenging to adapt to time-varying cooling loads and complex operating conditions. This limitation can result in suboptimal chiller operation, such as frequent on–off cycling or sustained operation outside of high-efficiency regions, further increasing energy consumption in HVAC systems [3].

Furthermore, MBC method lack adaptability in the presence of equipment degradation and are hampered by computing overhead and modeling mistakes, especially in large-scale and aging HVAC systems. With the continuous development of big data technologies and computational capability, reinforcement learning (RL), as a data-driven method, has increasingly become a mainstream way for HVAC control policy optimization [4]. It learns optimal control policies through agent–environment interactions, which endows RL with considerable adaptability to address problems like as equipment aging and environmental variability. Despite these advantages, an unavoidable problem in RL is its low sample efficiency [5]. Training an RL agent from scratch typically requires millions of interaction steps, resulting in prohibitively long training cycles—particularly for policy optimization tasks in the HVAC domain—which may take several months or even years, and incur extremely high deployment costs. Therefore, in real-world problems, it is vital to build efficient and cost-effective controllers that can strike a compromise between adaptability and practical feasibility.

In recent years, with the development of large language models (LLMs), they have displayed extraordinary logical, cognitive, and intellectual capacities [6,7]. At the same time, they have shown a deep comprehension of numerous complicated concepts, exhibited extremely detailed reasoning abilities, and can access information repositories across various aspects, such as medical, economics, and social media. This excellent information integration aptitude and good reasoning ability are precisely what make LLMs significant professionals in the HVAC business. In 2023, the Microsoft Asia team created a method for using LLM to manage HVAC systems (contextual learning control), which displayed performance comparable to or better than the reinforcement learning PPO method [8]. Notably, it involves no training and depends purely on the LLM's innate reasoning capabilities, suggesting that LLM can effectively learn to regulate HVAC equipment through prompting. However, this method still exhibits several limitations. In particular, its lack of selective learning ability prevents the LLM from accurately distinguishing inferior samples, resulting in decisions that do not conform to basic logic [9]. While parameter-efficient fine-tuning methods can address such problems, fine-tuning LLMs with billions of parameters is frequently prohibitively expensive, and if employing API-based LLMs, fine-tuning is impossible [10]. A more user-friendly method is to apply LLM with Retrieval-Augmented Generation (RAG) technology, which differs from fine-tuned LLM in that it does not require large-scale tweaks to the model's parameters. Instead, it enables the LLM to gain domain-specific knowledge through external information injection through an external retrieval module [11]. This method is more cost-effective and without requiring any training steps, it has the extra benefit of traceable information sources, which helps alleviate hallucinatory problems commonly observed in standalone LLMs. It is necessary to emphasize that the advantages of Retrieval-Augmented Generation (RAG) can only be fully realized when supported by a high-quality and well-structured domain knowledge base.

Therefore, this study focuses on the optimization of chiller operating methods in HVAC systems and presents a novel framework named the Cooperative LLM-based Control Framework (Co-LLM). It incorporates a domain knowledge base encompassing chiller operating policies, ASHRAE standards, and expert knowledge, and use Retrieval-Augmented Generation (RAG) to provide dynamic matching between environmental states and control policies. By using the contextual reasoning capabilities of large language models (LLMs), Co-LLM performs adaptive calibration of sensitive sub-policy without requiring explicit physical modeling. To further improve policy refining, this work proposes the RAG-tuned Exploration algorithm, which utilizes action space discretization and a selective recalibration method to optimize sensitive sub-policy.

The main contributions of this paper can be summarized as follows:

- We propose a Retrieval-Augmented Cooperative LLM-based Control Framework (Co-LLM) to enable real-time HVAC load optimization. The framework aims to minimize energy consumption while simultaneously ensuring indoor thermal comfort.
- We design a selective-recalibration mechanism that fine-tunes only sensitive sub-policies and integrate it with a RAG-tuned exploration policy for fast, targeted retrieval that uses the domain knowledge base for support.
- Extensive simulations on a real hospital chiller plant show that Co-LLM consistently achieves superior energy savings, comfort, and overall COP compared with strong baseline method, underscoring its effectiveness. Our code is publicly available for future research use.[1]

[1] Code repository: https://github.com/lgy163/collm.

2. Relate work

2.1. Traditional control methods

The traditional methods used to control chillers include Rule-Based Control (RBC) and Model-Based Control (MBC). RBC is centered around a manually designed rule base, with PID control being a typical example. It controls the chiller output using the linear combination of three components: Proportional (P), Integral (I), and Derivative (D). Sanama et al. applied an enhanced PID to improve the performance of the chilled water fan-coil unit in controlling temperature, humidity, and carbon dioxide levels [12]. However, RBC relies on expert expertise, and when there are too many input aspects, it can quickly lead to the "curse of dimensionality" with the number of rules expanding exponentially with the number of variables. Moreover, under dynamic loads, PID is prone to overshoot and lacks the ability to manage multi-objective constraints [13]. MBC, on the other hand, explicitly anticipates system behavior through physics-based or data-driven models and optimizes control actions under restrictions. Gao et al. suggested a model-based method that produced good performance in addressing Building Energy System (BES) optimization problems [14]. Hu et al. devised a two-layer model predictive control algorithm to increase the performance of air conditioning systems [15]. In another work, Gao et al. suggested a Tube-based model control technique for setting system temperature and fan control signals [16]. Although RBC and MBC have been widely adopted and achieved considerable success in practical engineering, RBC still strongly relies on expert experience and lacks adaptability to new environmental changes. MBC, on the other hand, relies on the accuracy of the model, and when equipment ages, the loss of model accuracy becomes unavoidable.

2.2. Reinforcement learning control methods

To overcome the inherent limitations of traditional control methods, the academic community has recently begun to bring reinforcement learning (RL) into the subject of HVAC control [17]. Extensive literature indicates RL can overcome the static assumptions and model dependencies of RBC control and model-based control (MBC), exhibiting higher adaptability under complex, time-varying situations in HVAC systems. Yu et al. noted in their research that many RL algorithms have been used to HVAC optimization, addressing diverse control tasks [18]. Chen et al. proposed a method using Q-learning to improve the window system of HVAC, considerably decreasing energy consumption and improving comfort [19]. However, Q-learning is limited by the expansion of state–action dimensions, making it difficult to directly tackle continuous high-dimensional control problems such as chiller units. He et al. proposed a model-free predictive control method for chiller units based on DQN and LSTM networks, exhibiting performance comparable to model predictive control [3]. Fu et al. introduced a novel framework called ED-MDP and a method called ED-DQN for optimizing multi-zone HVAC systems, achieving state-of-the-art performance [20]. With the increasing requirement for modeling continuous action spaces, the Deterministic Policy Gradient (DDPG) set of algorithms has been introduced into chiller control. Yang et al. introduced an enhanced DDPG-based algorithm to find the ideal operating variables of chiller systems under different loads, therefore optimizing their energy usage [21]. DDPG enhances sample efficiency through the use of an actor-critic architecture with target networks and an experience replay buffer. However, it is highly sensitive to noise and prone to convergence to suboptimal policies due to function approximation errors and policy instability. Wang et al. effectively improved performance by applying the SAC method in HVAC systems in office cooling-dominated environments [22]. Bai et al. proposed an advanced PPO method that decouples adversarial long short-term memory and near-term policy optimization, significantly decreasing energy consumption and improving comfort in a disrupted experimental [23]. Perera and Kamalaruban noted in their research that Actor-Critic methods achieved a 10% improvement in building energy management and control problems [24]. Compared to classic RBC and MBC systems, RL does not require modeling of chillers and instead approximates optimal control through trial-and-error learning, which can decrease costs to some extent. However, contemporary RL algorithms generally suffer from low sample efficiency and high training costs.

2.3. Large language model control methods

In recent years, large language models (LLMs) have emerged as an essential tool in intelligent decision-making scenarios [25–27]. Lei et al. directly applied the pre-trained GPT-4 directly to HVAC control and leveraged expert demonstrations and historical interactions to automatically screen for high-quality polices achieving cross-task transfer with nearly zero training cost [8]. However, the deployment of LLMs in industrial domains still faces notable limitations. In particular, the lack of domain-specific knowledge coverage and the tendency to generate hallucinated information hinder their safe application, especially when no correction mechanisms are available. To address this, Xiao et al. proposed a multi-agent framework incorporating retrieval-augmented generation (RAG) to dynamically inject building energy optimization knowledge. This method substantially reduced the manual cost of handling unstructured data while enhancing the level of automation [28]. Although LLMs have demonstrated remarkable potential in reasoning-based decision-making and knowledge extraction, current applications in HVAC control remain at the paradigm exploration stage. In particular, there is a lack of research into LLM-based knowledge integration and controllable reasoning in HVAC system control. How to fully unleash the potential of new technologies such as LLM and RAG in industrial intelligent control scenarios remains a key challenge that current research must address.

In this paper, we propose a framework: the Retrieval-Augmented Cooperative LLM Control Framework (Co-LLM), tailored for real-time energy-efficient control of chiller systems. This framework first performs retrieval from historical operational data, ASHRAE standards, and expert experience to select high-quality policy that best match current operating conditions. This step significantly

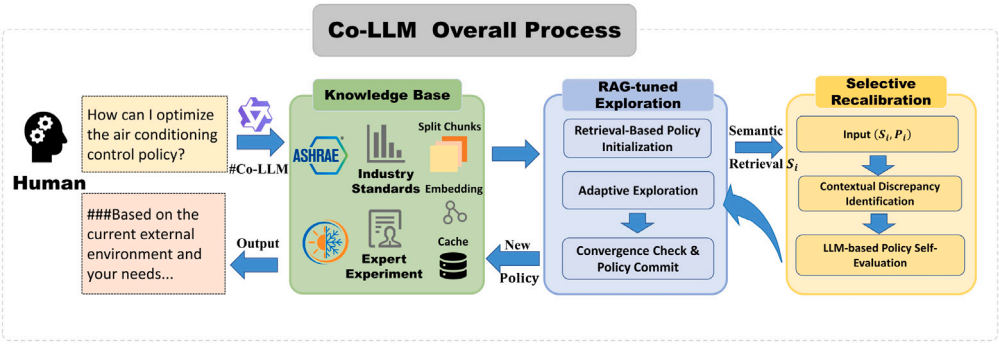


Fig. 1. Overall architecture of the proposed Co-LLM chiller-plant control framework.

reduces the hallucination probability inherent in pure LLM methods and ensures that the policy generation adheres to industry constraints. Then, the LLM performs reasoning and selective recalibration to iteratively fine-tune the sensitive sub-policies, enabling rapid convergence and dynamic adaptation. After each iteration, the validated optimum is stored in the knowledge-base cache for future use. Simulation results demonstrate that the proposed framework achieves lower energy consumption, improved comfort, and suppressed policy oscillations, all without requiring a precise physical model.

3. Methodology

The proposed Co-LLM is an innovative large language model that optimizes control polices using engineering data and expert knowledge, particularly utilized to optimize the control polices for chiller in HVAC systems. As indicated in Fig. 1, Co-LLM consists of three components: the chillercontrol knowledge base, selective recalibration, and RAG-tuned Exploration. When entering the control process, the retrieval-based initialization is first engaged, which obtains the policy segment most comparable to the present operating conditions from the database'cache as the warm-start policy. The process then proceeds to selective recalibration (SRC), fine correction for sub-policy where significant changes were detected. Finally, the framework performs performance evaluation during the RAG-tuned Exploration phase, storing the optimized policeinto the cache for quick reuse in similar future scenarios.

3.1. Chiller knowledge base

Similar to the original work on Retrieval-Augmented Generation (RAG), this study also utilizes the RAG framework to retrieve and integrate knowledge relating to chiller unit control, aiding in the generation of more specific control polices. Unlike general knowledge retrieval in open domains, this study focuses on chiller unit policy optimization, constructing a high-precision knowledge base (Chiller Knowledge Base) that effectively extracts and utilizes industrial site expert experience, maintenance logs, and equipment manuals, seamlessly integrating it into the Co-LLM framework.

In real-world industrial applications, the main challenges faced by the chiller knowledge base lie not only in the scattered information and diverse forms of papers and equipment manuals, but also in the difficulty in collecting and selecting valid expert knowledge from field experts. Some essential data usual exists in unstructured or implicit forms (e.g., verbal statements from maintenance staff, unit commissioning notes), making it particularly challenging to extract truly useful content. Through RAG, we are able to inject high-value content into the context window of the large language model (LLM), bridging the gap between industrial scenarios and pre-trained model knowledge.

3.1.1. Data preprocessing

In order to systematically integrate dispersed expert experience equipment information, we first extensively collected documents and data from multiple sources, recording all raw data sets as D_{raw}, which includes the following categories: $D_{raw} = \{D_1, D_2, D_3, \ldots, D_m\}$, where, m denotes the number of documents. The documents include vendor manuals, industry standards, and expert cases, with significant heterogeneity in their sources and formats. To ensure data consistency, we define a data cleaning function ϕ. In this process, we perform standard NLP operations, such as denoising, deduplication, punctuation normalization, sentence segmentation, and tokenization, and logically divide the document content to create an intermediate corpus D_i' for further processing: $D_i' = \Phi\left(D_i\right), i = 1, 2, \ldots, m$ The processed document collection is denoted as: $\mathcal{D}' = \{D_1', D_2', \ldots, D_m'\}$. Next, we perform semantic splitting on each document D_i' to obtain a collection of semantic fragments $d_1^i, d_2^i, \ldots, d_n^i$, for each document, which are stored as tuples containing key information. For example, a segment of an operation record can be represented as:

$$d_j^i = (timesamp = 6/3\ 18:35:00, \ldots, T_chws = 8, T_wet = 15.45)$$

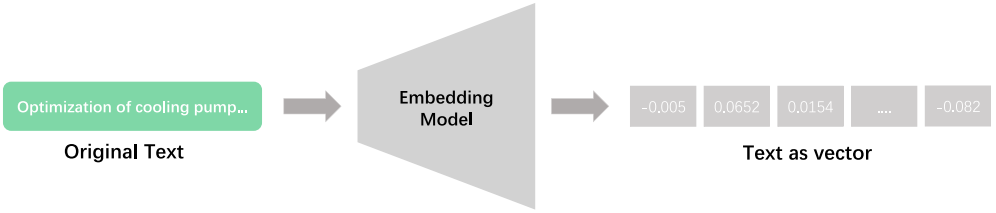


Fig. 2. Example of semantic text embedding for chiller-plant domain knowledge.

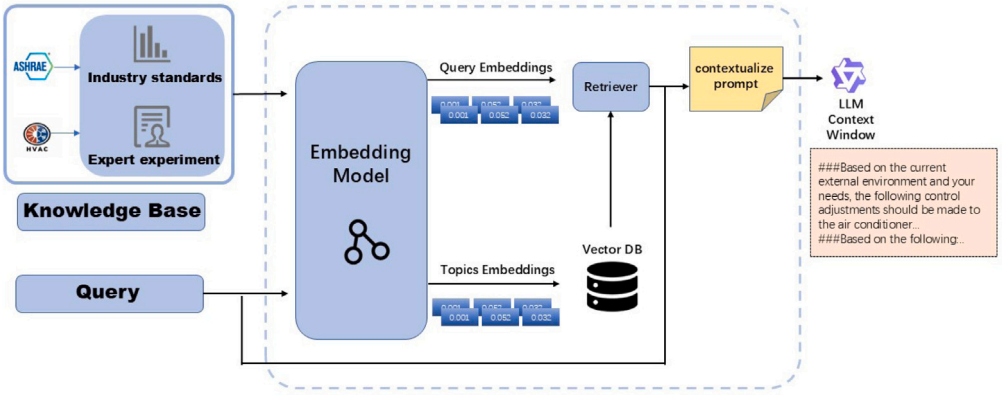


Fig. 3. RAG processing pipeline: knowledge retrieval, embedding, and prompt augmentation.

3.1.2. Text embedding and information retrieval

To convert the preprocessed text data into vector representations for semantic similarity calculation, we use the Text-Embedding-v3 model for text embedding. This model is specifically designed to convert text into high-dimensional vectors, preserving deep semantic information and supporting large-scale semantic search and similarity computation. Compared to traditional word embedding models, Text-Embedding-v3 is better at capturing contextual information and implicit semantic relationships in long texts. Specifically, the embedding process is shown in Fig. 2

We input each text fragment d_j^i into the embedding model to generate a corresponding semantic embedding vector $\boldsymbol{d}_j^i$. Similar text fragments will have smaller distances in the vector space. During retrieval, we use a cosine similarity-based Eq. (1) to calculate the similarity between the query vector $\boldsymbol{q}$ and all semantic embedding vectors $\boldsymbol{d}_j^i$ in the knowledge base:

$$\mathrm{sim}(\boldsymbol{q}, \boldsymbol{d}_j^i) = \frac{\boldsymbol{q} \cdot \boldsymbol{d}_j^i}{\|\boldsymbol{q}\| \left\|\boldsymbol{d}_j^i\right\|} \tag{1}$$

By calculating the similarity, the retriever selects the document fragments most relevant to the query context and integrates these fragments into the LLM's context window to generate more accurate replies. As shown in Fig. 3, our embedding model starts by vectorizing expert experience and other content. These embeddings are subsequently stored in a vector database, which as a searchable repository for retrieving relevant data. During a query, text fragments are first vectorized and embedded into the vector space, and then the most relevant fragments are retrieved and ranked. The top-ranked text fragments are added to the LLM's context window, providing relevant information to the model and enabling it to give more precise replies.

3.2. Selective recalibration

During the continuous operation of chiller systems, many working conditions exhibit only minor and localized fluctuations, such as slight fluctuations in the outdoor wet bulb temperature or short-term slight fluctuations in the cooling water pump flow. If a large language model (LLM) is prompted to perform full-scale reasoning for every minor fluctuation, two main issues arise. First, inference resources are wasted: when queries differ only slightly in wording yet pursue the same control objective, invoking the full reasoning process for each results in repeated generation under similar contexts, leading to unnecessary computational and temporal overhead. Second, maintaining policy consistency and coherence becomes difficult; because responses in different rounds are based on local prompts without a mechanism to compare them with previous outputs, the generated control policies are prone to inconsistency or conflict, thereby undermining the executability of the optimization policy.

Frequent occurrences of multi-turn, semantically similar queries pose significant challenges in terms of efficiency and consistency if high-quality existing responses cannot be effectively identified and reused. To address this problem, this study proposes a method

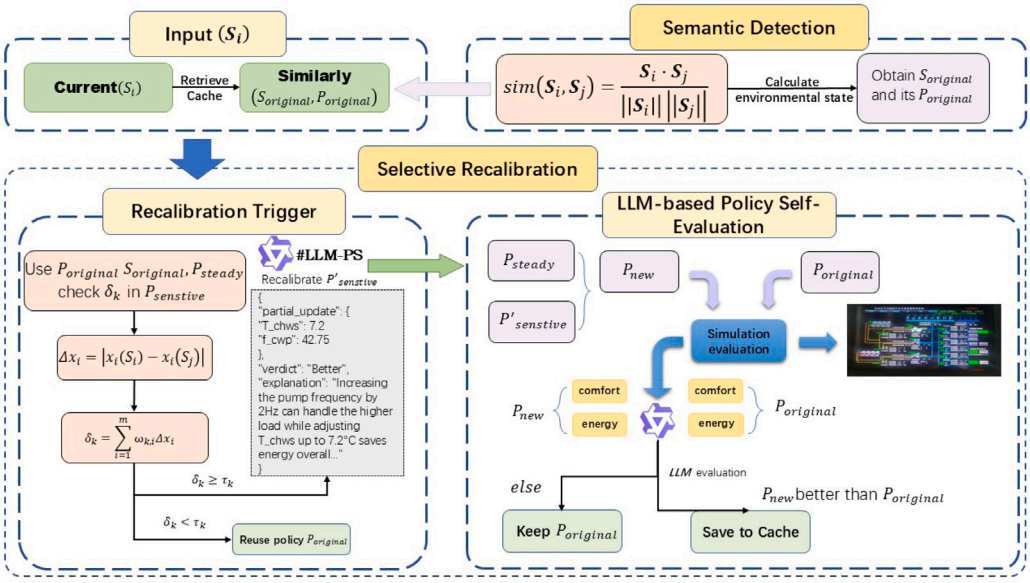


Fig. 4. Selective Recalibration workflow within the Co-LLM framework.

called Selective Recalibration, built upon the original multi-turn interaction framework. Let (S_i, P_i) denote the current pair of environmental state S_i and its corresponding initial policy P_i, and (S_j, P_j) represent a cached state-policy sample pair stored in the knowledge base'cache. It is assumed that a similar environmental state S_j has appeared in a previous round or earlier time window, and the associated policy P_j has been previously verified to perform well under that condition.

3.2.1. Local recalibration methods

In industrial chiller control, certain key sub-policy often exhibit only small-amplitude, time-varying; under such environmental state, applying a holistic policy-optimization method wastes computational resources, because most of the existing policy remains valid despite these slight environmental changes yet is still discarded and recomputed during global re-optimization. As illustrated in Fig. 4 SRC addresses the problem by concentrating on those sensitive sub-policy that are significantly affected by environmental variations and, with the aid of contextual-difference detection and LLM-driven policy evaluation, performing "local corrections" solely on components displaying substantial discrepancies while leaving steady sub-policy untouched. Let P denote the set of policy, partitioned as: $P = P_{\text{steady}} \cup P_{\text{sensitive}}, P_{\text{steady}} \cap P_{\text{sensitive}} = \varnothing$ where P_{steady} comprises insensitive sub-policy to minor environmental changes or requiring only minimal tuning—such as the number of cooling towers or the chiller on/off control logic, and $P_{\text{sensitive}}$ contains sub-policy highly responsive to environmental or load fluctuations, such as cooling-water-pump frequency and chilled-water supply temperature. During the recalibration stage, we therefore adapt only the sub-policy P'sensitive $\subseteq P$sensitive affected by the current conditions, while keeping P_{steady} unchanged wherever possible so as to reduce computational overhead and preserve the effectiveness of the existing policy.

3.2.2. Recalibration trigger

In SRC, it is necessary to determine whether the current environmental state and a cached environmental state are sufficiently similar, and whether the external features associated with sensitive sub-policies have undergone significant changes. This decision governs whether local variable recalibration is required. To formalize this process, for each sensitive sub-variable $s_k \in P_{\text{sensitive}}$, we associate a set of external features $x_1, \ldots, x_m$. For each feature x_i, we define the absolute change between the current environmental state S_i and the cached environmental state S_j as:

$$\Delta x_i = \left| x_i(S_i) - x_i(S_j) \right| \tag{2}$$

We then compute a parameter deviation metric δ_k for the sensitive sub-variable s_k as a weighted sum of all feature differences:

$$\delta_k = \sum_{i=1}^{m} \omega_{k,i} \Delta x_i \tag{3}$$

where $\omega_{k,i}$ denotes the relative weight of feature x_i in determining the behavior of s_k, set based on prior expert knowledge. A larger value of δ_k indicates that s_k is more significantly influenced by the environmental change. If $\delta_k \geq \tau_k$, where τ_k is a significance threshold for change, then s_k is considered to have changed substantially under the current condition relative to the cached condition, and must be added to the recalibration list for localized adjustment. We denote the recalibrated sub-policy of sensitive

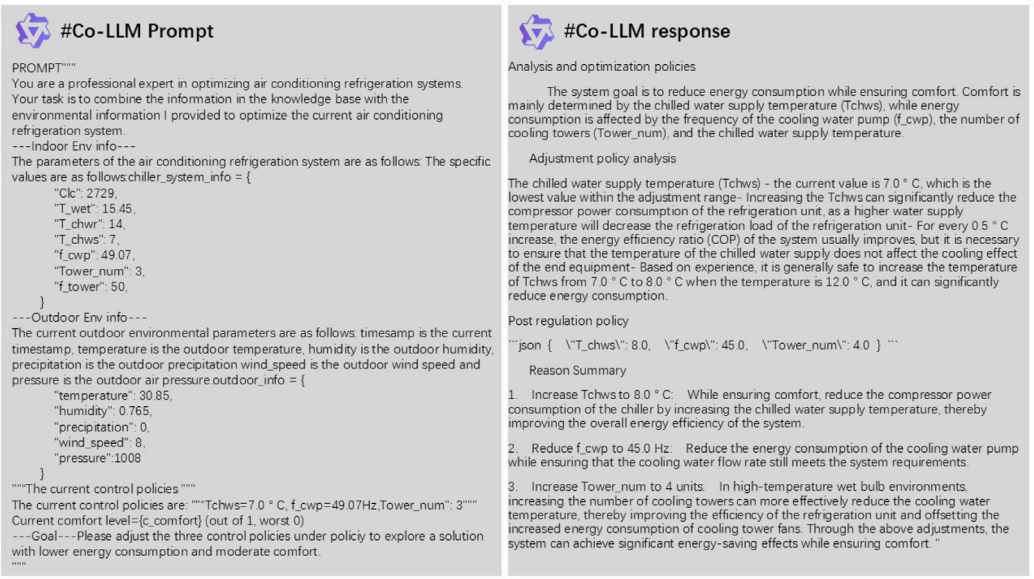


Fig. 5. Prompt interaction for LLM-based policy self-evaluation: prompt engineering examples, LLM response examples.

sub-policy as: $P'_{\text{sensitive}} = \text{Recalibrate}\left(P_{\text{sensitive}}, S_i, S_j\right)$, where, Recalibrate() refers to the process of leveraging a large language model to fine-tune the values of selected sensitive sub-policy based on the current input environmental state S_i, while referring to the original cached policy. If $\delta_k < \tau_k$, then the environmental influence on s_k is deemed insignificant, and the original policy value from P_{original} may be safely reused without modification.

3.2.3. LLM-based policy self-evaluation

After the local correction of sensitive sub-policy is completed, it is necessary to compare the recalibrated policy with the original one to determine whether the recalibrated version is worth adopting. To support this, LLM-based evaluation is introduced into the SRC workflow to ensure that the newly generated policy exhibits significant improvements in both performance and feasibility.

Once the recalibrated sensitive sub-policy $P'_{sensitive}$ have been updated, they are merged with the unchanged steady sub-policy P_{steady} to form the candidate policy: P_{new}. This candidate policy P_{new} is then submitted to the simulation environment to obtain evaluation metrics such as energy consumption and thermal comfort. These metrics are subsequently passed to the LLM for evaluation (Fig. 5). The LLM evaluates whether the new policy outperforms the original one across multiple dimensions. Fig. 5 illustrates the standardized prompt-response template: the upper panel shows the structured prompt that concatenates indoor and outdoor environment state, the current control policy, and an explicit optimization goal; the lower panel illustrates the model's structured reply, including rationale, step-wise adjustment analysis, the JSON-formatted policy, and a concise reason summary. This structured prompt-response format maintains the model's reasoning visible and allows its recommendations to be executed without any post-processing. If the overall performance difference is insignificant $M_{new} - M_{best} < 0.1\%$ (where the metric is defined in Section 3.3), the system retains the original policy and proceeds either with multi-turn dialog refinement or broader search, thus avoiding potential performance degradation caused by unnecessary local adjustments. Conversely, if the LLM determines that P_{new} yields better outcomes than the original, it accepts the recalibrated policy and stores it in the cache as a reusable state-policy pair: $\left(S_i, P_{new}\right)$ This allows for efficient reuse when similar operating conditions are encountered in the future.

3.3. RAG-tuned exploration

In the process of policy optimization centered around LLM, although it is theoretically possible to invoke the model an unlimited number of times to develop multi-round optimization policies, this does not imply that every adjustment is viable or practically significant. If the system relies only on repeated LLM calls, it may fall into an unending trial-and-error loop, where excessive resource consumption fails to provide effective convergence. Moreover, some output policies may only perform well under short-term or localized conditions, and excessive rounds of optimization may further amplify uncertainty, making it difficult to determine whether an acceptable near-optimal policy has been reached. To address these issues, this paper building on a systematic review and synthesis of existing research [29–31] and introduces a composite evaluation metric that integrates energy consumption and thermal comfort, which is used to measure the effectiveness of a policy and further proposes an RAG-tuned exploration algorithm to balance continued exploration and iterative termination in the multi-round policy optimization process.

3.3.1. Energy-comfort evaluation metric

(1) **Energy metric**

The total power consumption P_{total} is defined as the sum of the power consumption of three types of equipment: cooling towers, chillers, and cooling pumps:

$$P_{\text{total}} = \sum_{i,j,k=1}^{n} (P_{\text{tower},i} + P_{\text{chiller},j} + P_{\text{c_pump},k})$$

where $P_{\text{tower},i}$ denotes the power consumption of the ith cooling tower, $P_{\text{chiller},j}$ is the power consumption of the jth chiller, and $P_{\text{c_pump},k}$ is the power consumption of the kth cooling pump.

(2) **Comfort metric**

The overall comfort score C_{comfort} is defined as a weighted sum of indoor and outdoor comfort scores:

$$C_{\text{comfort}} = \omega_{\text{indoor}} C_{\text{indoor}} + \omega_{\text{outdoor}} C_{\text{outdoor}}$$

The indoor comfort score is defined as:

$$C_{\text{indoor}} = e^{-\alpha_{\text{chwr}}(T_{\text{chwr}} - T_{\text{opt}})^2}$$

where T_{chwr} is the chilled water return temperature (° C), T_{opt} is the optimal reference temperature set to 12°C, and $\alpha_{\text{chwr}} > 0$ is the exponential decay coefficient, set to 0.03 in this study. The outdoor comfort score C_{outdoor} is calculated based on five environmental variables: temperature, humidity, precipitation, wind speed, and atmospheric pressure. Each variable is evaluated through a non-linear scoring function f_x, as follows: For temperature, the sub-score is defined as:

$$f_{\text{temp}} = e^{-\beta_{\text{temp}}(T_{\text{out}} - T_{\text{ref}})^2}$$

where T_{out} is the outdoor temperature (° C), $\beta_{\text{temp}} = 0.02$, and the reference temperature is $T_{\text{ref}} = 25$°C.
For humidity, the sub-score is given by:

$$f_{\text{hum}} = e^{-\beta_{\text{hum}}(\text{RH}_{\text{out}} - \text{RH}_{\text{ref}})^2}$$

where RH_{out} is the outdoor relative humidity, $\beta_{\text{hum}} = 0.015$, and $\text{RH}_{\text{ref}} = 50\%$.
For precipitation, the sub-score is defined as:

$$f_{\text{precip}} = e^{-\beta_{\text{precip}} \cdot \text{precip}}$$

with $\beta_{\text{precip}} = 0.1$, so heavier rainfall rapidly decreases the score.
For wind speed, the sub-score is defined piecewise as:

$$f_{\text{wind}}(w) = \begin{cases} e^{-\gamma_1 (w - \omega_{\text{opt}})^2}, & w \le \omega_{\text{opt}} \\ e^{-\gamma_2 (w - \omega_{\text{opt}})^2}, & w > \omega_{\text{opt}} \end{cases}$$

where the optimal wind speed is $\omega_{\text{opt}} = 2.0$ m/s, with decay rates $\gamma_1 = 0.05$ and $\gamma_2 = 0.3$.
For atmospheric pressure, the sub-score is expressed as:

$$f_{\text{pressure}} = e^{-\beta_{\text{pressure}}(\text{pressure} - p_{\text{ref}})^2}$$

where $\beta_{\text{pressure}} = 0.005$, and the reference pressure is $p_{\text{ref}} = 1013$ hPa.
Let the five sub-scores be denoted as:

$$\{s_1, s_2, s_3, s_4, s_5\} = \{f_{\text{temp}}, f_{\text{hum}}, f_{\text{precip}}, f_{\text{wind}}, f_{\text{pressure}}\}$$

To reflect the relative importance of each variable, a set of weights $\{w_1, w_2, w_3, w_4, w_5\}$ is applied in the form of a weighted geometric mean:

$$F_{\text{outdoor}} = \prod_{i=1}^{5} (s_i)^{w_i}$$

The weights used in this study are $w_{\text{temp}} = 0.3$, $w_{\text{hum}} = 0.2$, $w_{\text{precip}} = 0.25$, $w_{\text{wind}} = 0.15$, and $w_{\text{pressure}} = 0.1$.
To dynamically adjust the relative importance of indoor vs. outdoor comfort in varying weather, particularly under different precipitation levels, a logistic precipitation-based weight function is defined:

$$w'_{\text{precip}} = \frac{1}{1 + e^{-\delta(\text{precip} - p_{\text{th}})}}$$

where $\delta = 0.5$ and $p_{\text{th}} = 5$ mm/h.
The final dynamic weights are calculated as:

$$\omega_{\text{outdoor}} = \min\left(0.6,\ 0.3 + 0.3 w'_{\text{precip}}\right), \quad \omega_{\text{indoor}} = 1 - \omega_{\text{outdoor}}$$

The final comfort score is computed as:

$$C_{\text{comfort}} = \omega_{\text{indoor}} \cdot e^{-\alpha_{\text{chwr}}(T_{\text{chwr}}-T_{\text{opt}})^2} + \omega_{\text{outdoor}} \cdot \prod_{i=1}^{5} (s_i)^{w_i}$$

(3) **Normalization and Composite metric**

Before computing the composite metric, both energy consumption and comfort are normalized.

$$P_i = \max\left\{0, \frac{P_{\max} - P_{\text{total}}}{P_{\max} - P_{\min}}\right\}$$

$$C_i = \max\{0, \min[1, C_{\text{comfort}}]\}$$

Here, $P_{\max}$ and $P_{\min}$ are determined from the past year of historical data, with $P_{\max}$ taken as the 95th-percentile value and $P_{\min}$ as the 5th-percentile value.

The final performance metric is then calculated as:

$$M_t = \alpha P_i + \beta C_i$$

with weighting coefficients $\alpha = 0.5$ and $\beta = 0.5$.

We further assess the robustness of the proposed comfort metric to weight perturbations. A one-at-a-time sensitivity sweep (with re-normalization to keep $\sum_i w_i = 1$) is conducted under Sunny/Rainy/Windy regimes; the resulting trends are stable and regime-consistent, indicating that the metric is not dominated by arbitrary weight choices. Full details are provided in Appendix C.2.

3.3.2. Algorithm

Algorithm 1: RAG-tuned Exploration

Input: S_i: current environment state; R_{init}: initial iteration budget; R_{limit}: maximum iteration budget; ΔR: budget expansion step; θ_{imp}: improvement threshold; $N_{\max}$: maximum stagnation threshold; $\xi(P)$: evaluation function returning performance metric M

Output: P_{best}; M^*: best policy and metric

$R_{\max} \leftarrow R_{\text{init}}$; $T \leftarrow 0$; $N \leftarrow 0$; $(S_j, P_j) \leftarrow$ Retrieval – Based_Policy_Initialization(S_i) ; // fetch most similar state-policy pair from cache

$P_{\text{steady}} \leftarrow \{ p \in P \mid \text{isSteady}(p)\}$; $P_{\text{sensitive}} \leftarrow P \setminus P_{\text{steady}}$; // Split the policy into steady sub-policy, sensitive sub-policy

if *Recalibration_Trigger* **then**

  while $T < R_{\max}$ **do**

    $P'_{sensitive} \leftarrow \text{Recalibrate}(P_{sensitive}, S_i, S_j)$; // Recalibrate the current sensitive sub-policy

    $P_{new} \leftarrow P'_{sensitive} + P_{steady}$

    $M_{new} \leftarrow \xi(\text{P}_{\text{new}})$; $M_t \leftarrow \xi(\text{P}_{\text{j}})$

    if $(M_{new} > M_t)$ ***and*** $|M_{new} - M_t| > \theta_{imp}$ **then**

      $M^* \leftarrow M_{new}$; // update best metric

      $P_{best} \leftarrow P_{new}$; // update best policy

      $N \leftarrow 0$; // reset no improvement count

    else

      $N \leftarrow N + 1$; // increment no improvement count

    if $N > N_{\max}$ **then**

      return P_{best}, M^*

    $T \leftarrow T + 1$; // increment iteration counter

    if $T = R_{\max}$ ***and*** $R_{\max} < R_{limit}$ **then**

      $R_{\max} \leftarrow \min(R_{\max} + \Delta R,\ R_{\text{limit}})$; // increase budget to explore further

  return P_{best}, M^* **and** save (S_j, P_{best}) to cache

else

  return P_j

Based on the previous metrics, we propose an algorithm presented in Algorithm 1 that dynamically balances the policy boundary between continued exploration and iterative termination, thereby preventing wasted computational resources and ineffective iterations. The algorithm begins by taking the current environment state (S_i) and retrieving from the knowledge base'cache the most similar state-policy pair (S_j, P_j). The policy P_j is splitted into a steady sub-policy P_{steady} and a sensitive sub-policy $P_{\text{sensitive}}$.

Table 1
Specifications of HVAC equipment.

Device	Quantity	Specifications
Screw chiller	1	Cooling capacity: 1404 kW; Power: 258 kW; Chilled water supply temperature: 8°C.
Centrifugal chiller 1	1	Cooling capacity: 3024 kW; Power: 534.8 kW; Chilled water supply temperature: 8°C.
Centrifugal chiller 2	1	Cooling capacity: 2813 kW; Power: 573.25 kW; Chilled water supply temperature: 8°C.
Chilled water pump	6	Flow rate: 200 m^3/h; Power: 22.2 kW; Frequency: 50 Hz (variable speed).
Condenser water pump	3	Pumps 1–2: Flow rate 800 m^3/h; Power 90 kW; variable-speed. Pump 3: Flow rate 375 m^3/h; Power 45 kW; variable-speed.
Cooling tower	8	Towers 1–4: Flow rate 1400 m^3/h; Power 11 kW; fixed-speed. Towers 5–8: Flow rate 1500 m^3/h; Power 15 kW; fixed-speed.

If the predefined *Recalibration_Trigger* is satisfied, the algorithm enters an exploration loop: at each iteration the sensitive sub-policy is locally recalibrated, yielding $P'_{\text{sensitive}}$, which is then merged with P_{steady} to form a candidate policy P_{new}. The evaluation function ξ computes the composite metrics M_{new} for P_{new} and M_t for (S_j, P_j). Whenever $M_{\text{new}} > M_t$ and the improvement satisfies $|M_{\text{new}} - M_t| > \theta_{\text{imp}}$, the best metric M^* and best policy P_{best} are updated and the stagnation counter N is reset; otherwise N is incremented. If N exceeds the maximum stagnation threshold N_{patience}, the loop terminates early and returns (P_{best}, M^*). After each pass the iteration counter T is incremented; when $T = R_{\max}$, provided that $R_{\max} < R_{\text{limit}}$, the budget is expanded to $R_{\max} \leftarrow \min(R_{\max} + \Delta R, R_{\text{limit}})$. Upon loop completion the pair (S_j, P_{best}) is written back to cache for rapid reuse under similar future conditions, whereas if recalibration is not triggered, the original policy P_j is return directly.

The algorithm continuously and dynamically adjusts and refines the chiller control policy through an iterative, self-evaluating closed-loop process. At each iteration, policy effectively incorporate domain-specific knowledge provided by RAG. The algorithm allows deeper iterations while performance continues improving, thus fully exploring potential policies. Conversely, when policy performance stabilizes, it terminates promptly to avoid unnecessary computational overhead. A glossary of acronyms and mathematical symbols is provided in Appendix A.1 and A.2. For reproducibility, the full parameter settings are reported in Appendix B.1 and B.2

4. Construction of simulation environment

In this section, we discuss the evaluation of our proposed method, providing a detailed description of the experimental setup, the conducted tests, and the obtained results.

4.1. Case study

The case study presented in this work is based on the actual HVAC system of a hospital located in Taizhou, China (28.6644°N, 121.4208°E) is located in Zhejiang Province, where summers are hot and humid, and winters are relatively mild. The total floor area of the building is approximately 114,000 m^2. The cooling system in this case consists of three chillers: one screw chiller with a cooling capacity of 1404 kW, and two centrifugal chillers with capacities of 3024.6 kW and 2813 kW, respectively. The primary chilled-water system is equipped with six chilled-water pumps (each 22.2 kW), while the cooling-water system comprises two cooling-water pumps rated at 90 kW each, one pump at 45 kW, and four cooling towers at 11 kW each. The structure of the cooling system is illustrated in Fig. 7, and the detailed equipment configuration and parameters are summarized in Table 1. All cooling-water pumps support variable-frequency control, enabling them to adjust their speed to meet cooling-water demand under varying load conditions, thereby enhancing heat exchange efficiency. It is noteworthy, however, that the cooling towers operate at a fixed frequency with fans maintaining a constant rotational speed.

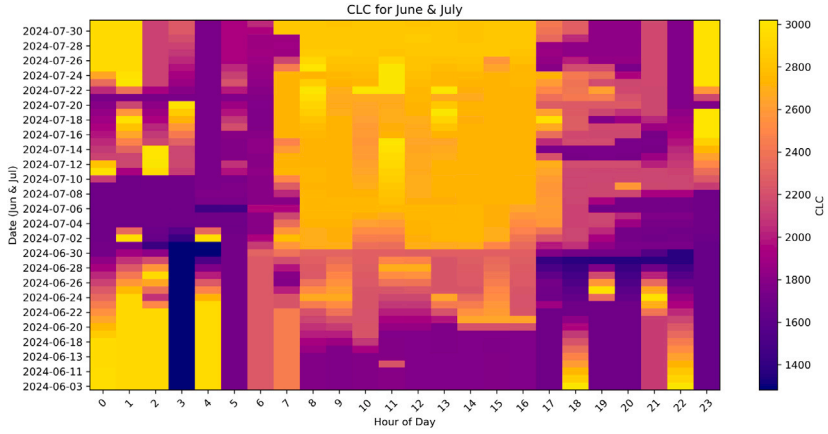
The validation procedure of this study is based on actual operational data. Specifically, we collected hourly cooling load data from the hospital's chiller units, as well as local outdoor temperature data during the peak cooling season from June to July 2024. The variation in cooling load is illustrated in Fig. 6(a), and the corresponding temperature changes are shown in Fig. 6(b).

In practical chiller policy optimization, directly depending on frequent interactions between LLMs and real equipment can cause the models to issue unstable control actions during early exploration and require extended periods to converge to usable operating policies a method clearly impractical for engineering deployment. To avoid the risks and costs of lengthy trials on real systems, many researchers instead conduct training and testing in simulation environments, which has proven to greatly enhance controller performance [32].

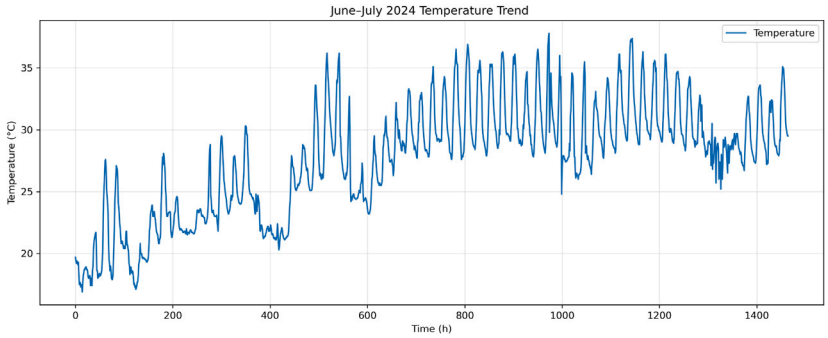
5. Experimental results and discussion

5.1. Experimental setup

In this section, we compare the proposed Co-LLM with three classical methods commonly applied to chiller control problems: RBC control, model-based control, and DQN-based control. To comprehensively evaluate the effectiveness of the proposed method, we focus on three distinct aspects: overall performance, energy-saving performance, and operational performance.



(a) Changes in cooling load in the case environment.



(b) Outdoor temperature changes in the case environment.

Fig. 6. Cooling load and outdoor temperature changes in the case environment — June to July.

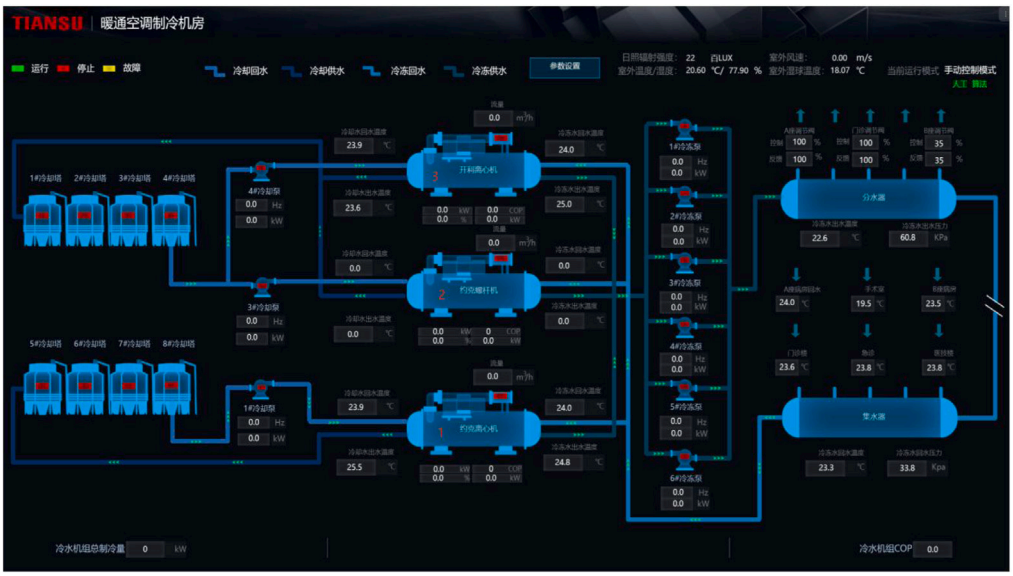


Fig. 7. Schematic of the cooling system, showing the core components and their interactions. The No. 2 York screw machine and No. 3 carrier centrifuge are the primary units responsible for supplying cooling water to Cooling Towers No. 1 through No. 4 on the left. The cooling towers return the water through Cooling Pumps No. 3 and No. 4. On the right, chilled water is supplied to the six chilled water pumps (No. 1 to No. 6), which then deliver chilled water to the water separator, receiving the chilled return water from the water collector.

Table 2
Co-LLM main parameter configuration.

Parameter	Value
LLM model	Qwen series (primarily Qwen Max)
RAG framework	LightRAG
Evaluation model	Qwen Max
Initial policy reuse score threshold	9
Initial rounds R_{init}	5
Maximum rounds R_{limit}	20
No-improvement limit $P_{\text{no_imp}}$	5
Improvement threshold θ_{imp}	0.01
Cache size	10 MB

5.1.1. Knowledge base construction

In constructing the RAG knowledge base specifically for chiller policy optimization, this study extensively incorporates authoritative literature and expert experiences across multiple domains, aiming to enhance the knowledge base's professional depth and practical applicability. Specifically, core literature and standards published by ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) are first included, such as the *ASHRAE Handbook-HVAC Applications*, *ASHRAE Handbook-HVAC Systems and Equipment*, *ASHRAE Guideline 36: High-Performance Sequences of Operation for HVAC Systems*, and *ANSI/ASHRAE Standard 55: Thermal Environmental Conditions for Human Occupancy*. These resources offer comprehensive industry best practices concerning energy-saving policies and high-performance operating modes.

Furthermore, the knowledge base integrates extensive expert experiences and practical engineering cases, including but not limited to the chiller group-control scheme of Shanghai Metro Line 9 and energy-saving retrofit design methodologies applied in hospital HVAC systems. Such practical cases provide substantial empirical data and optimization policies from various perspectives, including multi-unit operation control and measured energy-saving outcomes, directly reflecting how dynamic regulation policies can effectively balance energy consumption and occupant comfort under varying load conditions and operational requirements. The integration of these authoritative references and industry experiences into the RAG knowledge base not only supplies valuable prior information for this study but also renders the chiller policy optimization process more convincing and operationally feasible. At the policy optimization stage, the RAG framework can rapidly retrieve and leverage relevant knowledge, thus enabling the LLM to generate control schemes that align closely with practical engineering demands, significantly enhancing energy utilization efficiency while ensuring indoor comfort.

5.1.2. Parameter setting for Co-LLM

The core LLM adopted in this study is Qwen Max [33], which demonstrates superior performance in complex instruction understanding, multi-round policy optimization, and domain-specific knowledge integration. With hundreds of billions of parameters, Qwen Max maintains strong reasoning capabilities while significantly reducing memory consumption through dynamic computational graph optimization, enabling efficient deployment even on a single RTX 4090 GPU. As a prominent representative among open-source Chinese LLMs, the Qwen series exhibits outstanding performance in industrial understanding tasks, ranking within the top three on the CLUE benchmark, and supports flexible lightweight fine-tuning policies. In this study, we employ the LightRAG framework to construct the retrieval-augmented system, whose novel hierarchical indexing structure reduces retrieval latency to below 50 ms, thus satisfying real-time chiller control requirements [34]. Table 2 summarizes the key experimental environment configurations in detail: Co-LLM includes several practical hyper-parameters (Table 2) controlling policy reuse and early stopping. When deploying to a new site, we recommend calibrating the initial policy reuse score threshold using a short held-out period to balance cache reuse versus recalibration, and choosing the improvement threshold θ_{imp} according to the desired trade-off between policy quality and iteration (LLM-call) budget. We conduct a targeted one-at-a-time sensitivity analysis on the current case scenario by sweeping (i) the reuse threshold in $\{7, 8, 9, 10\}$ with θ_{imp} fixed to 0.01, and (ii) θ_{imp} in $\{0.005, 0.01, 0.015, 0.02\}$ with the reuse threshold fixed to 9, while keeping all other parameters unchanged. We report monthly energy, mean comfort, and the average number of optimization rounds, showing that the proposed method remains stable in a broad neighborhood around the default configuration and that these parameters mainly modulate the reuse frequency and iteration cost. Detailed results are reported in Appendix C.1 (varying s_{reuse}) and Appendix C.2 (varying θ_{imp}). The results indicate that energy and comfort are stable around the default configuration ($s_{\text{reuse}} = 9$, $\theta_{\text{imp}} = 0.01$), whereas the average number of optimization rounds is more sensitive, reflecting the main deployment trade-off between performance and iteration cost. For s_{reuse}, increasing the threshold slightly improves energy consumption (June: 71,528 kWh → 71,413 kWh, a reduction of 0.16%; July: 101,754 kWh → 100,916 kWh, a reduction of 0.83%) and comfort (nearly unchanged), but increases the iteration cost substantially (7.4 → 10.3 rounds from $s_{\text{reuse}} = 9$ to 10). For θ_{imp}, lowering the threshold increases rounds (7.4 → 8.6) with only marginal performance gain, while overly large thresholds reduce rounds (down to 6.2) but can lead to a noticeable energy penalty in July (up to +1.1%). Overall, the default setting provides a favorable balance between performance and iteration budget, and the sensitivity trends offer practical guidance for adapting the method under different computational constraints.

Table 3
Hyperparameter settings for the comparative DQN method.

Hyperparameter	Value
batch_size	128
learning_rate	0.0003
gamma	0.98
buffer_size	10 000
exploration_initial_eps	1.0
exploration_final_eps	0.2
net_arch	[256, 128, 64]
Episodes	200

Table 4
Energy consumption comparison for June and July.

Control method	June (kW h)	July (kW h)
Model-based	69 762	99 710
RBC	72 883	101 609
Co-LLM	71 456	101 132
DQN	71 256	101 372

5.1.3. Comparison methods

In this experiment, we compare four control methods: model-based control, pure DQN-based control, RBC control, and the proposed Co-LLM-based control. The model-based control method serves primarily as a baseline to assess the performance gap between our method and an optimal control method. The DQN-based control method is used to benchmark against traditional reinforcement learning methods. Meanwhile, the RBC control method provides a practical reference, enabling comparison between real-world energy consumption and the proposed Co-LLM method, thereby highlighting the energy-saving potential of our control method. The core components and implementation details of each method are summarized in Table 2. The cooling load profiles and meteorological data employed in this experiment originate from actual measurements recorded at Taizhou Hospital from June to July 2024, ensuring the engineering authenticity of our evaluation scenario. All methods operate within the cooling system simulation environment described in Section 4.1, and the output policies from each method are evaluated using the same energy-consumption and comfort models.

(1) Model-based Control: This method acquires current environmental state information $[\mathrm{CLc}, T_{\mathrm{wet}}, \ldots]$, and at each simulation step, systematically explores all possible control actions: tower_num $= \{0, 1, 2, 3, 4\}$ $T_{\mathrm{chws}} = \{7, 8, 9, \ldots, 12\}$ $f_{\mathrm{cwp}} = \{32.0, 33.0, \ldots, 50.0\}$ and computes the energy consumption for each action accordingly.
(2) DQN-based Control: This method is model-free deep reinforcement learning methods with the same environmental space and discrete action space as model-based control. The parameter settings are shown in Table 3.
(3) RBC Control: This method relies on manually controlled chiller operation data from domain experts in a practical setting. The energy consumption data was obtained through an on-site investigation of the hospital's HVAC system.

5.2. Experimental result analysis

5.2.1. Algorithm performance analysis

Fig. 8 shows the convergence trajectories of energy consumption and comfort over four randomly selected operational sub-figures. In each plot, the horizontal axis S_1- S_n represents successive iteration rounds, while the vertical axes depict the composite energy consumption P_{total} and the composite comfort score $C_{comfort}$ (see Section 3.3.1 for details) respectively. Across all sub-figures, the energy curve exhibits a monotonic decrease, whereas the comfort curve rises steadily and smoothly—with no sign of oscillatory behavior. These results confirm that the Co-LLM framework, via its retrieval-based initialization and SRC, successfully suppresses the policy-fluctuation issues typically seen in pure-LLM methods, thereby delivering high-quality control policies in real time. In terms of iteration efficiency, only 7.4 iterations are required to achieve 95% of the final energy savings, and the comfort metric approaches its upper bound within the first seven iterations. In the subsequent 2–3 fine-tuning rounds, both metrics change by less than 0.5%, indicating that the algorithm remains robust under small perturbations in operating conditions. This performance gain arises from three mutually reinforcing design choices: RAG-based Retrieval Initialization provides high-quality warm-start policies, significantly reducing the cost-intensive exploration phase. SRC constrains the search space to only the sensitive sub-policies, greatly lowering exploration complexity. LLM Self-Evaluation directs the optimization process away from inefficient search routes, hence limiting cyclical and divergent policy tendencies.

5.2.2. Energy consumption analysis

Table 4 presents the specific energy consumption of the different control methods. In June, Co-LLM consumed 71 456 kWh, representing a 1.9% reduction compared to the RBC policy (72 883 kWh) and within 0.3% of the DQN-based policy (71 256 kWh).

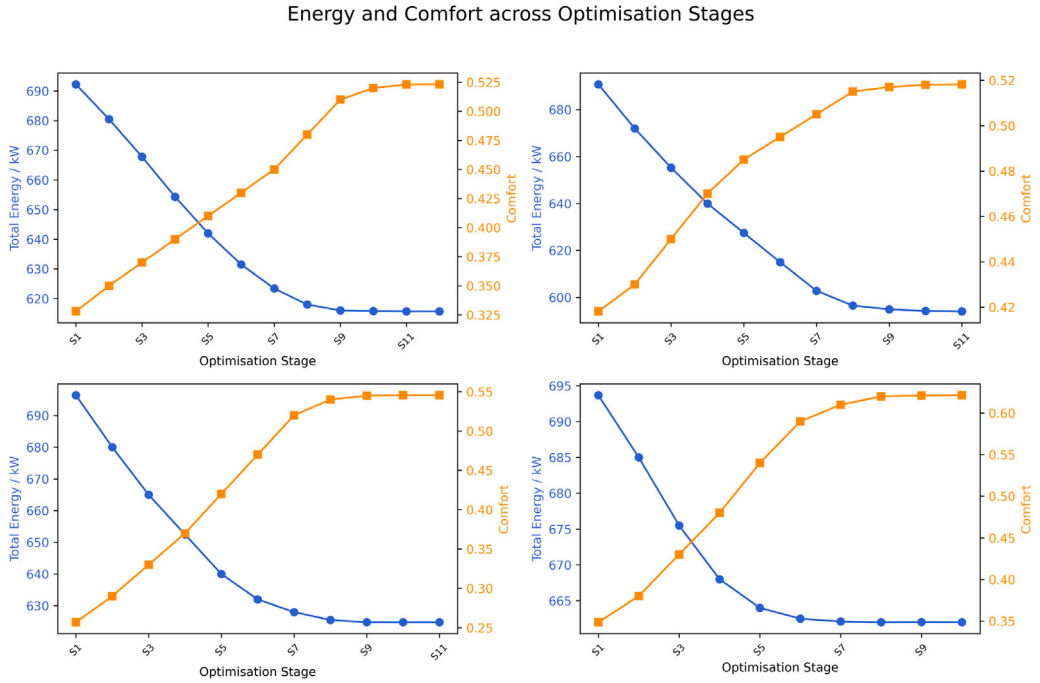


Fig. 8. Convergence of total energy and comfort metrics across five randomly selected operation windows.

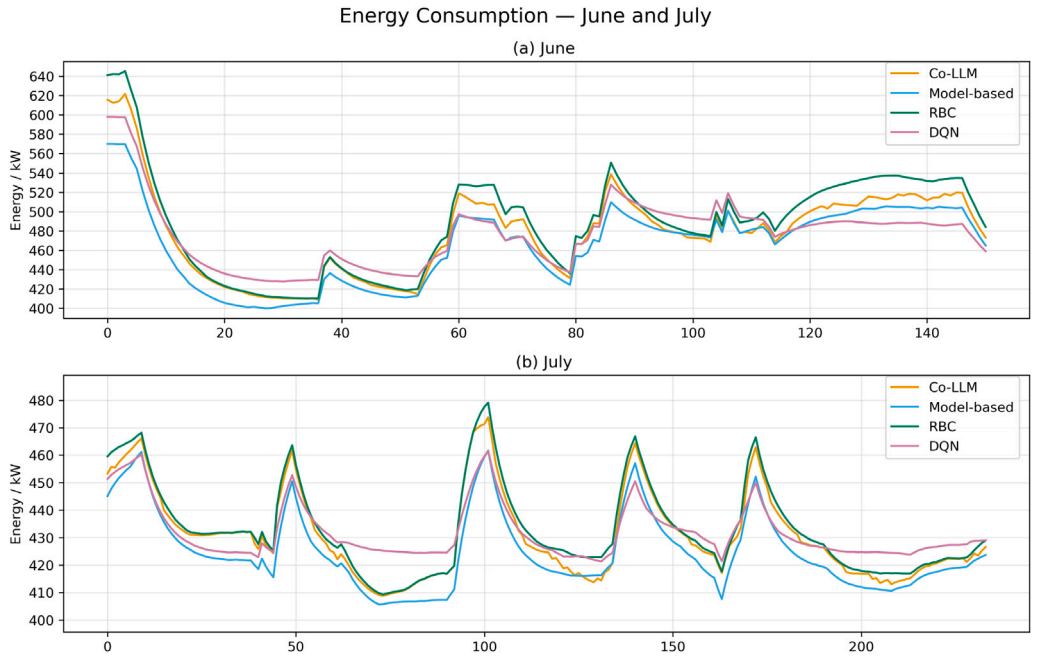


Fig. 9. Energy consumption comparison for four methods (Co-LLM, Model-based, RBC, DQN) on the June (a) and July (b) samples.

During July, Co-LLM's consumption (101 132 kWh) was further reduced by 0.5% relative to the RBC policy (101 609 kWh) and achieved a 1.4% energy-efficiency advantage over DQN (101 372 kWh).

Fig. 9 plots the energy consumption of the four control methods under each operating condition, clearly illustrating the performance gradient among the methods, the horizontal axis represents the index of operating condition, and the vertical axis represents energy consumption. Co-LLM consistently occupies the lowest-consumption band and steadily approaches the theoretical optimum across two months of independent load samples: compared with the exhaustive enumeration baseline, Co-LLM falls behind

Table 5
Comfort score comparison for June and July.

Control method	June (0–1)	July (0–1)
Model-based	0.621	0.660
RBC	0.648	0.681
Co-LLM	0.658	0.682
DQN	0.457	0.518

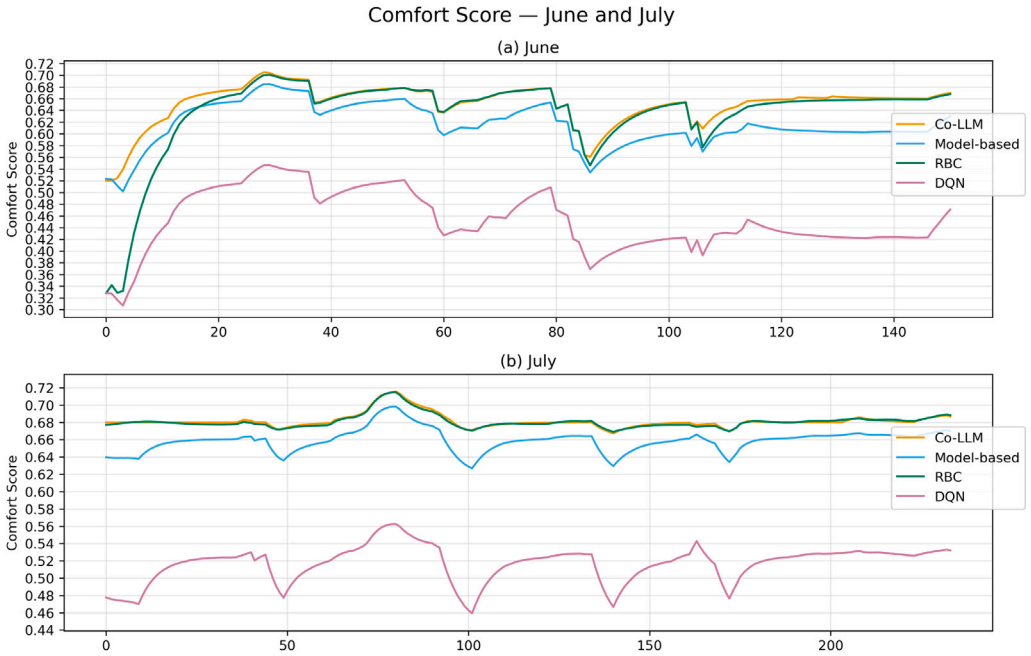


Fig. 10. Comfort score comparison for four methods (Co-LLM, Model-based,RBC, DQN) on the June (a) and July (b) samples.

by only 2.4% in June and narrows the gap to 0.5% in July. It is worth noting that, although the DQN control method achieves lower energy consumption than both Co-LLM and RBC under certain operating conditions, it does so at the cost of a pronounced reduction in comfort Fig. 10. These results validate the effectiveness of Retrieval-Based Policy Initialization and SRC: RAG supplies the LLM with high-quality warm-start policies, and the LLM then recalibrates sensitive sub-policies based on simulation feedback to approximate the optimal solution.

5.2.3. Comfort analysis

Table 5 presents the average comfort scores of the four control methods. Co-LLM achieved the highest scores of 0.658 and 0.682, representing improvements of approximately 1.5% and 0.15% over the RBC policy, and increases of 6.0% and 3.3% relative to model-based control. In contrast, the DQN-based method scored only 0.457 and 0.518, underperforming the other methods and falling below 0.40 in several operating conditions—an indication of its insufficient policy stability.

Fig. 10 depicts the temporal evolution of comfort scores under each operating condition for the four control methods. the horizontal axis represents the index of operating condition, and the vertical axis represents comfort score. The comfort curve for Co-LLM is not only smooth but also exceptionally compact, almost coinciding with the RBC policy and exhibiting only minor intersections at a few points. This demonstrates that, with Retrieval-based policy initialization and SRC, the LLM's reasoning accurately captures and reuses the nuanced details of expert policies. In contrast, the model-based method—while generally stable—deviates under certain extreme conditions due to action-space discretization and modeling approximations. The DQN method, lacking expert experiences guidance, suffers from blind-spot exploration in early stages, resulting in pronounced comfort fluctuations. By maintaining its energy-saving advantage, Co-LLM achieves both higher average comfort and significantly lower variance, underscoring its robust reliability for deployment in large-scale real-world buildings.

5.2.4. COP analysis

Table 6 reports the COP values of the four control methods. In June, Co-LLM achieved an average COP of 4.43, slightly surpassing RBC (4.36, +1.6%) while trailing model-based control (4.52, –2.0%) and DQN (4.46, –0.7%). Under heavier loads in July, its performance dropped to 4.15 (–6.3% from June), yet it still outperformed RBC (4.13, +0.5%), matched DQN (4.15), and remained

Table 6
Average Coefficient of Performance (COP) for June and July.

Control method	June (COP)	July (COP)
Model-based	4.52	4.21
RBC	4.36	4.13
Co-LLM	4.43	4.15
DQN	4.46	4.15

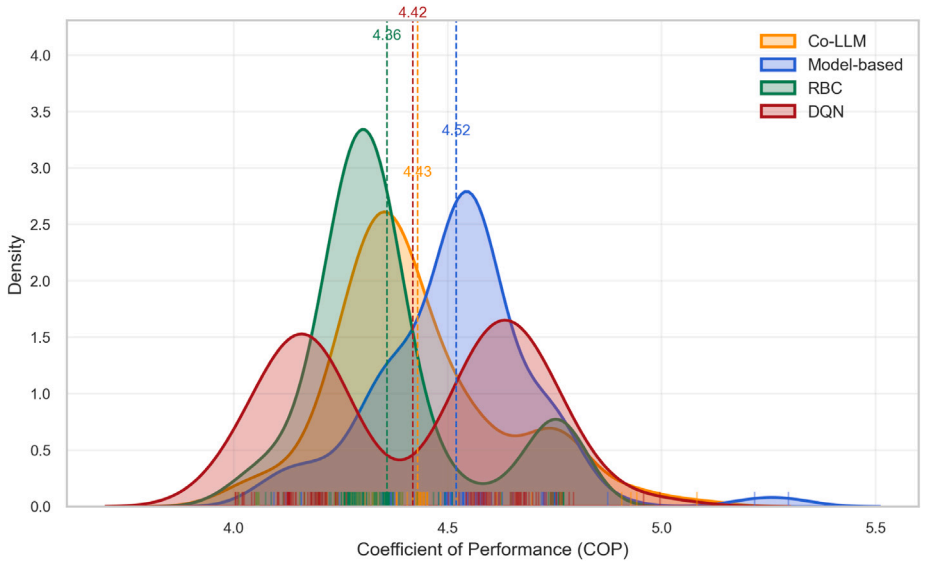


Fig. 11. Coefficient of Performance (COP) density distribution in four methods (Co-LLM, Model-based, RBC, DQN) on the June samples.

Table 7
Comparison of computational complexity and operational cost.

Metric	Co-LLM	RBC	MBC	DQN
Training time (min)	N/A	N/A	≈200	≈4.3
Inference latency (ms)	≈2000	≈0.01	≈2100	≈0.3
Memory usage (MB)	≈20	≈5	≈40	≈100

close to model-based control (4.21, –1.4%). Relative to the model-based method, the energy penalty was limited to 1.0% in June and 0.5% in July, highlighting Co-LLM's efficiency despite its simplified modeling.

Figs. 11 and 12 illustrate the COP distributions in July and June, respectively. As shown, the differences among the four control methods are relatively small in terms of efficiency, with Co-LLM maintaining competitive COP values comparable to model-based control and DQN, and consistently outperforming the rule-based method (RBC). On the surface, DQN appears to perform reasonably well in COP, showing values close to Co-LLM and even surpassing RBC.

However, the comfort results reveal a very different picture. Fig. 10 and Table 5 present the comfort scores for both months. While Co-LLM achieves the highest comfort levels (0.658 in June and 0.682 in July), closely followed by RBC and model-based control, DQN exhibits substantially lower comfort (0.457 in June and 0.518 in July). This sharp contrast indicates that although DQN maintains decent COP, it fails to deliver acceptable comfort, making it unsuitable for practical deployment. In contrast, Co-LLM achieves a favorable trade-off, combining high energy efficiency with superior comfort performance under varying load conditions.

5.2.5. Computational complexity and operational cost analysis

In addition to control performance, computational efficiency and operational cost are critical factors for the practical deployment of HVAC control strategies in large-scale buildings. Table 7 presents a quantitative comparison of the proposed Co-LLM and three representative baseline methods in terms of training time, inference-time computational complexity and latency, memory usage, and estimated computing cost.

As summarized in Table 7, Co-LLM does not require explicit offline training, similar to rule-based control (RBC), which greatly simplifies deployment and reduces maintenance burden compared to learning- or model-based approaches. In online operation, Co-LLM incurs an average inference latency of approximately 2000 ms per control step, which is on the same order as model-based control (MBC, ≈2100 ms) and substantially higher than RBC (≈0.01 ms) and DQN (≈0.3 ms). However, unlike RBC, Co-LLM provides adaptive decision-making rather than relying on fixed heuristics. In terms of memory footprint, Co-LLM requires around 20 MB,

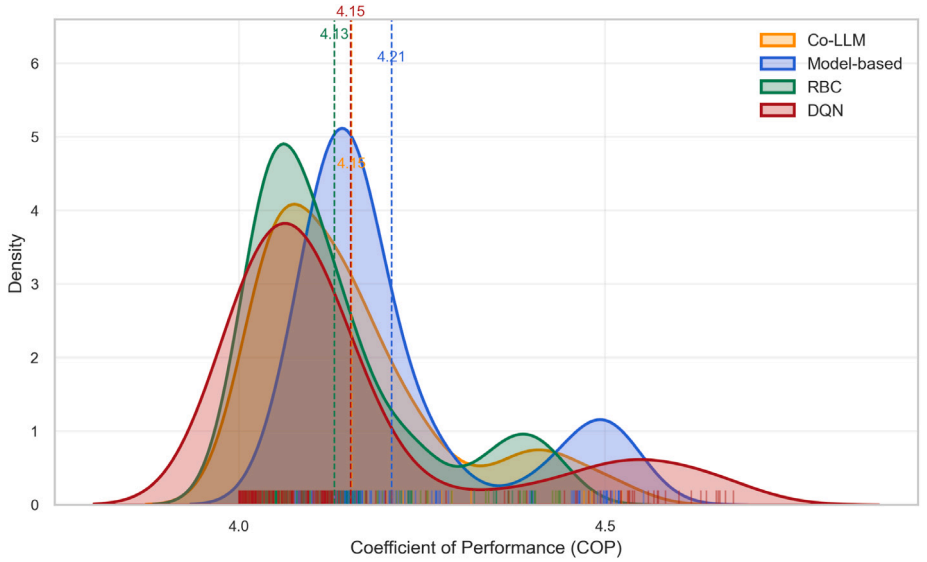


Fig. 12. Coefficient of Performance (COP) density distribution in four methods (Co-LLM, Model-based, RBC, DQN) on the July samples.

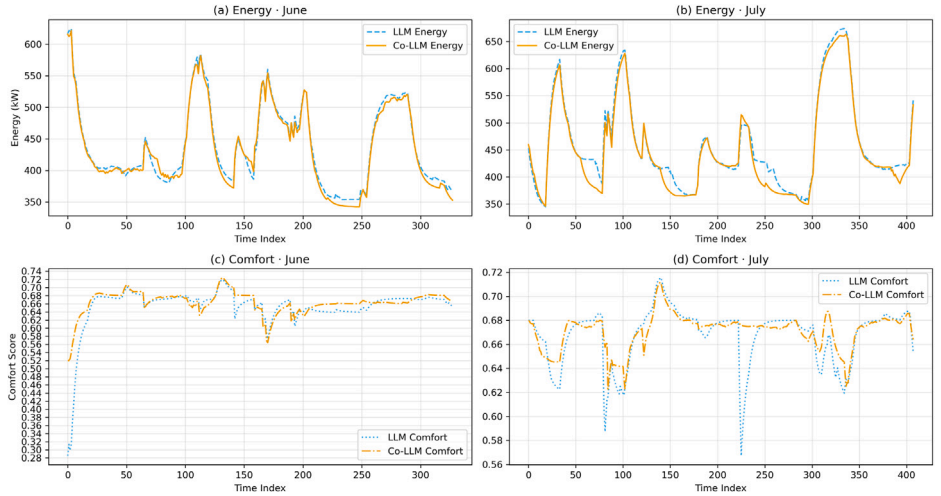


Fig. 13. Energy and comfort comparison of Co-LLM vs. LLM-only panels (a) and (b) plot the instantaneous energy consumption profiles for the pure LLM (dashed blue) and the retrieval-augmented Co-LLM (solid orange) over representative June and July operational windows, respectively. Panels (c) and (d) show the corresponding comfort scores on the same time axes.

which is considerably smaller than DQN (approximately 100 MB) and MBC (approximately 40 MB), while remaining higher than RBC (approximately 5 MB). Overall, these results indicate that Co-LLM offers a practical trade-off: it avoids costly offline training and maintains a relatively modest memory footprint, at the expense of higher per-step latency than lightweight baselines, making it a feasible option for real-world HVAC deployment where adaptability is needed under moderate computational constraints.

5.2.6. Ablation experiment

In the focused ablation test isolating the retrieval module, we compared the full Co-LLM against an otherwise identical LLM-only variant. Both methods used the same baseline policy initialization and were evaluated on identical operating points. Although their average energy curves nearly overlapped under most conditions, the retrieval-disabled variant exhibited significant extreme events shown in Fig. 13.

In June, the LLM-only control exhibited noticeable fluctuations in both energy consumption and comfort score, with occasional sharp drops in comfort below 0.6 and minor power spikes. These anomalies were significantly suppressed in Co-LLM, which maintained smoother energy trajectories and more stable comfort levels throughout the month. In July, under more extreme load conditions, the instability of the LLM-only method became more pronounced, with several deep dips in comfort and irregular energy

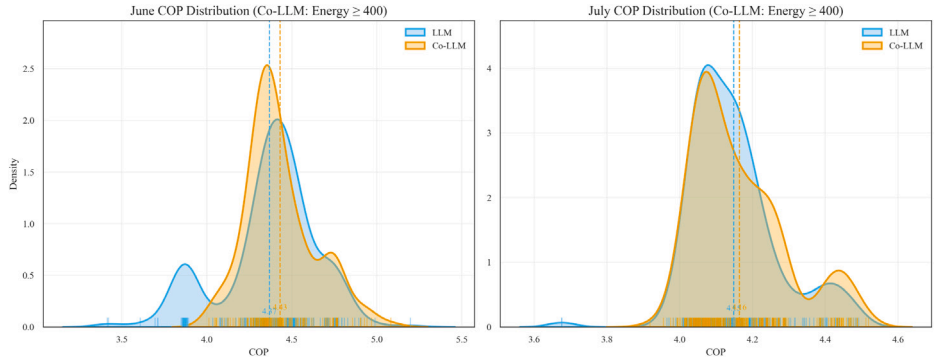


Fig. 14. Coefficient of Performance (COP) density distribution of Co-LLM vs. LLM-only in June and July.

Table 8
LLM backbone dependency analysis on June and July workloads. All settings except the backbone LLM are kept identical.

Backbone LLM	June		July	
	Energy (kWh)	Comfort	Energy (kWh)	Comfort
Doubao-seed-1.8	70 994	0.64	101 598	0.67
DeepSeek-v3	68 822	0.65	99 150	0.67
Kimi-k2	69 681	0.62	101 548	0.67
GLM-4.7	69 498	0.64	101 022	0.67
Co-LLM	71 456	0.66	101 132	0.68

peaks. In contrast, Co-LLM demonstrated superior robustness, maintaining a consistently higher average comfort level and avoiding severe performance degradation.

In the COP distribution Fig. 14, Co-LLM raised the mean COP from 4.37 to 4.42 in June and achieved a 1.27% improvement in July, while extending the high-efficiency tail beyond 4.30. Compared to the variant without RAG, these results highlight the critical role of retrieval augmentation: it not only boosts average performance but also suppresses extreme outliers. By leveraging domain knowledge, Co-LLM compresses the low-efficiency and high-risk regions, mitigating abnormal fluctuations. Under complex HVAC operating conditions, this leads to a more robust and energy-efficient control policy, thereby enhancing both reliability and energy savings.

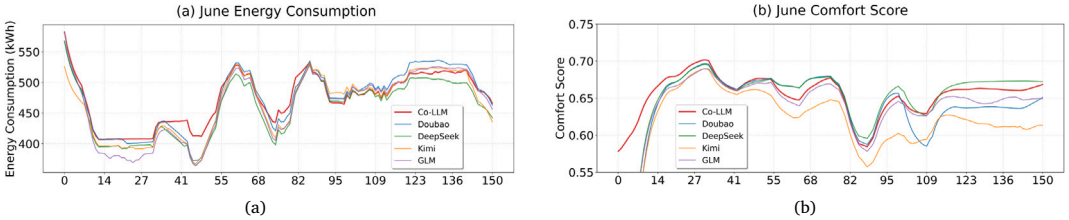
5.2.7. LLM dependency and reproducibility analysis

A key practical consideration is whether the proposed framework remains reproducible when the backbone LLM is changed. To examine this, we repeat the complete Co-LLM pipeline by replacing the backbone LLM with multiple alternatives, while keeping the rest of the system strictly identical: the same retrieval-augmented cache and initialization procedure, the same selective recalibration logic and action constraints, the same simulator and evaluation metrics, and the same prompt template and decoding configuration. Table 8 summarizes the quantitative results on the June and July workloads. Across different backbones, the comfort score remains within a narrow band (June: 0.62–0.65; July: 0.67–0.68), suggesting that the comfort outcomes produced by the proposed framework are consistent and not driven by a particular LLM choice. Energy consumption shows modest variations across backbones, which is expected because different LLMs may propose slightly different local adjustments even under the same constraints. Importantly, the overall conclusion remains unchanged: the Co-LLM pipeline consistently achieves competitive energy usage while maintaining high comfort.

Figs. 15, 16 further visualizes the energy and comfort outcomes under different backbones for June and July. The trends are broadly aligned across LLMs, and no single model exhibits systematically unstable comfort behavior. This stability is largely attributable to the design of Co-LLM: the backbone LLM operates within a constrained closed-loop procedure (retrieval warm start → local recalibration → simulation-based evaluation), which reduces sensitivity to model-specific phrasing and mitigates uncontrolled policy drift. Overall, these results indicate that the proposed method is not dependent on a single backbone LLM and that the reported findings are reproducible under multiple backbone choices.

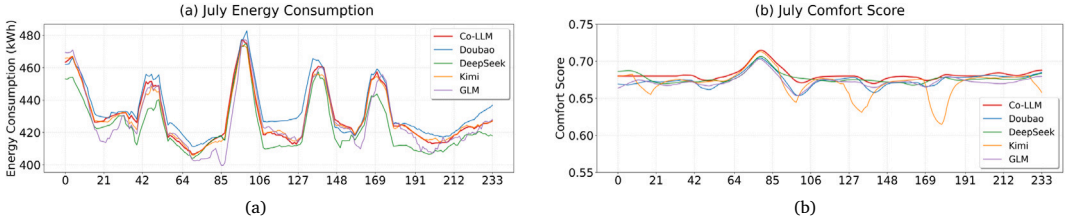
6. Conclusion and future work

This paper investigates method optimization for HVAC systems, focusing specifically on chiller control. To this end, we propose a Retrieval-Augmented Cooperative LLM-based control framework (Co-LLM). Co-LLM first builds a semantically vectorized knowledge



(a) (b)

Fig. 15. Backbone LLM dependency analysis on the June workload. The left subfigure (a) shows energy consumption, while the right subfigure (b) shows comfort levels. All settings, including system parameters and environmental conditions, remain identical except for the backbone LLM. The colored labels represent the following methods: red for Co-LLM, blue for Doubao, green for DeepSeek, yellow for Kimi, and purple for GLM. The analysis illustrates the contribution of the backbone LLM to the system's policy under the given workload conditions.



(a) (b)

Fig. 16. Backbone LLM dependency analysis on the July workload. The left subfigure (a) shows energy consumption, while the right subfigure (b) shows comfort levels. All settings, including system parameters and environmental conditions, remain identical except for the backbone LLM. The colored labels represent the following methods: red for Co-LLM, blue for Doubao, green for DeepSeek, yellow for Kimi, and purple for GLM. The analysis illustrates the contribution of the backbone LLM to the system's policy under the given workload conditions.

base that encodes ASHRAE standards, historical operating policies, and expert-log data as high-dimensional, searchable vectors representing chiller-dispatch expertise. At each control cycle, the framework employs retrieval-based initialization to select the prior policy most similar to the current environment condition; this warm-start policy is then locally refined by the large language model, which adjusts only those sub-policies sensitive to external disturbances (such as cooling-water-pump frequency) thereby avoiding the redundant overhead and policy drift that accompany global modifications. The updated policy is fed into a simulation environment in real time to obtain energy-consumption and comfort feedback; if the gain is insufficient, another refinement round is triggered to iteratively approaches a feasible optimum. To validate the effectiveness of the proposed method, we conducted an experimental case study on an actual hospital HVAC system, comparing four control methods in terms of energy use, comfort, and COP. The results show that Co-LLM achieves energy efficiency comparable to MBC method, delivers greater comfort than the RBC method, and markedly suppresses comfort drops. Its COP distribution shifts rightward, enlarging the high-efficiency region.

This paper confirm the feasibility and robustness of combining knowledge injection with large-model reasoning for industrial HVAC control, offering a new pathway for chiller-control optimization. However, several limitations remain. First, Co-LLM still depends on the reliability and output consistency of the underlying LLM; under rare, ambiguous, or borderline operating states, the model may generate suggestions that are syntactically valid yet operationally sub-optimal, and occasional run-to-run variability cannot be fully eliminated. Second, the effectiveness of retrieval-based initialization and selective recalibration depends on the coverage, freshness, and quality control of the knowledge base and cache; incomplete case coverage or outdated engineering notes may reduce warm-start quality and increase the number of refinement rounds. Third, our current empirical evidence is primarily drawn from a limited set of sites and a single seasonal regime, which constrains the extent to which we can characterize generalization across diverse climates and operating conditions. In future work, we will address these limitations in three corresponding directions: explore domain-specialized or domain-adapted LLM backbones (beyond general-purpose models) to improve instruction faithfulness and reduce occasional sub-optimal suggestions under rare or ambiguous operating states; develop automated knowledge-base maintenance and validation pipelines to continuously update domain documents and cached policies, detect retrieval drift, and prevent stale or low-quality references from degrading warm starts; and broaden evaluation across more sites and seasons—including stress tests under extreme weather and atypical load events—to better characterize generalization and to calibrate comfort objectives to site-specific requirements.

Table A.1
Glossary of acronyms.

Acronym	Definition
HVAC	Heating, Ventilation and Air Conditioning system
RBC	Rule-Based Control
MBC	Model-Based Control
RL	Reinforcement Learning
PID	Proportional–Integral–Derivative controller
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
Co-LLM	Cooperative Large Language Model (proposed framework)
SRC	Selective Recalibration (local policy adjustment)
COP	Coefficient of Performance
DQN	Deep Q-Network
DDPG	Deep Deterministic Policy Gradient
PPO	Proximal Policy Optimization
SAC	Soft Actor–Critic
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BES	Building Energy System
CLUE	Chinese Language Understanding Evaluation benchmark

CRediT authorship contribution statement

Guangyu Liu: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Yunzhe Wang:** Writing – review & editing, Validation, Investigation, Conceptualization. **You Lu:** Writing – review & editing, Validation. **Hongjie Wu:** Writing – review & editing, Validation, Supervision. **Ke Liu:** Writing – review & editing, Validation. **Lanhui Liu:** Writing – review & editing, Validation, Resources. **Jianping Chen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Qiming Fu:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

A.1. Glossary of acronyms

See Table A.1.

A.2. Mathematical symbols

See Table A.2.

Appendix B. Parameter settings for reproducibility

B.1. Algorithm hyper-parameters and trigger settings

See Table B.1.

B.2. Weights and thresholds used in selective recalibration

See Table B.2.

Table A.2
Primary mathematical symbols used in the paper.

Symbol	Description
P_{total}	Total electric power of towers, chillers, and pumps (kW)
C_{comfort}	Composite thermal-comfort score (0–1)
T_{chws}	Chilled-water supply temperature (°C)
T_{chwr}	Chilled-water return temperature (°C)
T_{out}	Outdoor air temperature (°C)
f_{cwp}	Cooling-water-pump frequency (Hz)
ω_{indoor}	Weight of indoor comfort in C_{comfort}
ω_{outdoor}	Weight of outdoor comfort in C_{comfort}
M_t	Composite performance metric at iteration t
M^*	Best-so-far composite metric
P_{steady}	Policy subset insensitive to small disturbances
$P_{\text{sensitive}}$	Policy subset requiring local recalibration
δ_k	Deviation score for sensitive variable s_k
θ_{imp}	Minimum improvement threshold for accepting new policy

Table B.1
Key hyper-parameters and trigger settings.

Parameter	Value	Description
R_{init}	5	Initial iteration budget of RAG-tuned exploration
R_{limit}	20	Maximum allowable iteration budget
ΔR	5	Budget-expansion step when improvement persists
$N_{\max}$	5	Early-stop patience (no-improvement limit)
θ_{imp}	0.01	Minimum metric gain to accept a new policy
`batch_size`	128	Mini-batch size in DQN training
`learning_rate`	3×10^{-4}	Adam learning rate in DQN
`gamma`	0.98	Discount factor in DQN
`buffer_size`	10 000	Replay-buffer capacity in DQN

Table B.2
Weights $\{\omega_{k,i}\}$ and thresholds τ_k.
Sources: Expert prior → empirical tuning on a held-out validation set (e.g., 2024-06 07).

s_k	Feature set (summary)	Weights $\{\omega_{k,i}\}$	τ_k
T_{chws}	$\{C_{lc}, T_{wet}, T_{chwr}, ws\}$	[0.40, 0.30, 0.20, 0.10]	0.35
f_{cwp}	$\{C_{lc}, T_{wet}, T_{chwr}, ws\}$	[0.20, 0.50, 0.20, 0.10]	0.45
Tower_*num*	$\{C_{lc}, T_{wet}, T_{chwr}\}$	[0.50, 0.30, 0.20]	0.30

Notes. (i) All features are z-score normalized before computing Δx_i in Eq. (2); weights are ℓ_1-normalized per s_k. (ii) Grid search for τ_k over $[0.20, 0.60]$ with step 0.05; (iii) **Rationale for** Tower_*num* **excluding wind speed:** in our mechanical-draft configuration, short-term wind perturbations are better absorbed by continuous fan-speed control; adding ws at the staging layer brought negligible gains.

Appendix C. Sensitivity analyses

C.1. Sensitivity to key Co-LLM hyper-parameters

This section reports one-at-a-time sensitivity results for key hyper-parameters while keeping all other settings fixed.

C.2. Sensitivity of the comfort metric to weight perturbations

We further evaluate the robustness of the comfort metric under one-at-a-time weight perturbations across representative weather regimes.

The outdoor comfort term aggregates five meteorological sub-scores (temperature, humidity, precipitation, wind speed, and pressure) via a weighted geometric mean. This design is intentionally non-compensatory: a severely adverse factor (e.g., extreme

Table C.1
Sensitivity to the initial policy reuse score threshold s_{reuse} with $\theta_{\text{imp}} = 0.01$. The default setting is $s_{\text{reuse}} = 9$.

Month	s_{reuse}	Energy (kWh)	Comfort	Avg rounds
June	7	71 528	0.657	6.5
June	8	71 490	0.658	6.8
June	9	71 456	0.658	7.4
June	10	71 413	0.658	10.3
July	7	101 754	0.680	6.5
July	8	101 680	0.681	6.8
July	9	101 132	0.682	7.4
July	10	100 916	0.683	10.3

Table C.2
Sensitivity to the improvement threshold θ_{imp} with $s_{\text{reuse}} = 9$. The default setting is $\theta_{\text{imp}} = 0.01$.

Month	θ_{imp}	Energy (kWh)	Comfort	Avg rounds
June	0.005	71 423	0.659	8.6
June	0.010	71 456	0.658	7.4
June	0.015	71 615	0.656	6.5
June	0.020	71 663	0.656	6.2
July	0.005	101 116	0.683	8.6
July	0.010	101 132	0.682	7.4
July	0.015	102 280	0.682	6.5
July	0.020	102 226	0.682	6.2

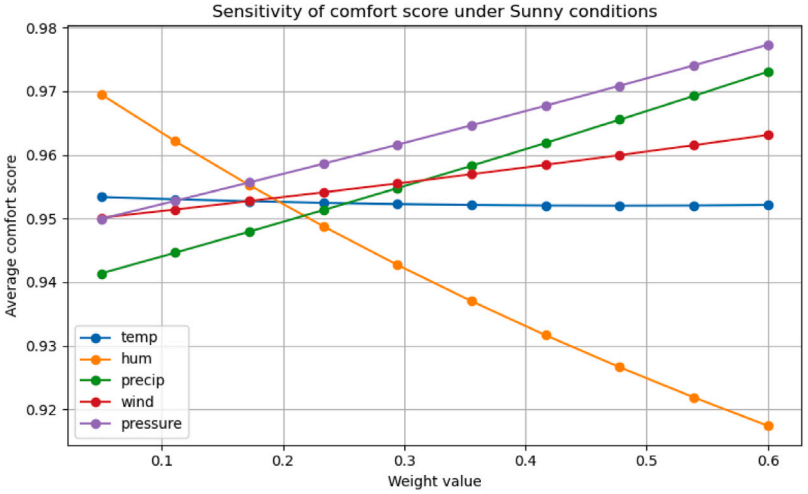


Fig. C.1. OAT sensitivity of the comfort score under Sunny conditions. One weight is varied while the remaining weights are re-normalized to keep $\sum_i w_i = 1$; the score is averaged over a multi-point meteorological grid.

humidity or heavy precipitation) cannot be fully offset by other favorable variables, which better reflects discomfort perception and avoids overly optimistic comfort estimates.

The nominal outdoor weights

$$(w_{\text{temp}}, w_{\text{hum}}, w_{\text{precip}}, w_{\text{wind}}, w_{\text{pressure}}) = (0.30, 0.20, 0.25, 0.15, 0.10)$$

were chosen as a balanced setting to reflect (i) the dominant impact of temperature/humidity on thermal discomfort and cooling-load stress, (ii) the direct penalty of precipitation events, and (iii) wind/pressure as secondary but non-negligible factors affecting outdoor discomfort and weather severity. Importantly, these weights are not tuned to any controller; they are fixed across all methods to ensure fair comparison.

To assess robustness, we perform a one-at-a-time sensitivity analysis: one weight is swept in $[0.05, 0.60]$ while the remaining weights are proportionally re-normalized to keep $\sum_i w_i = 1$. We evaluate three representative weather regimes (Sunny/Rainy/Windy) using multi-point sampling of meteorological conditions. Figs. C.1–C.3 show stable and interpretable trends: the comfort score is most sensitive to humidity weight under rainy/humid conditions and to wind weight under windy conditions, while temperature

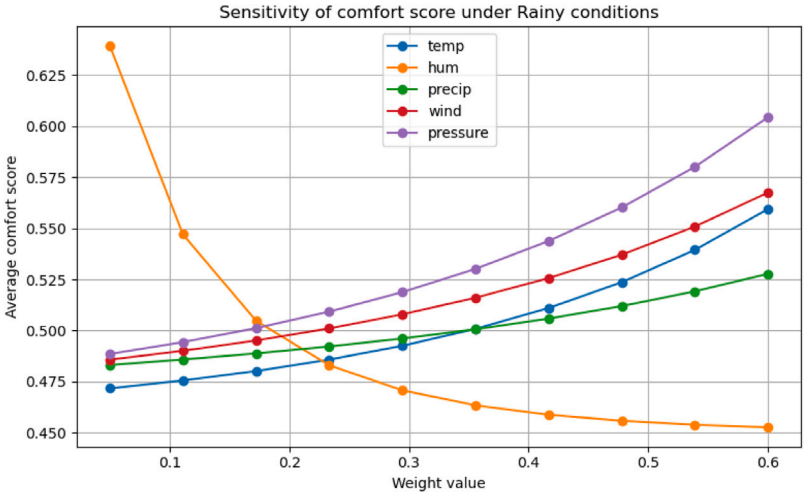


Fig. C.2. OAT sensitivity of the comfort score under Rainy conditions. One weight is varied while the remaining weights are re-normalized to keep $\sum_i w_i = 1$; the score is averaged over a multi-point meteorological grid.

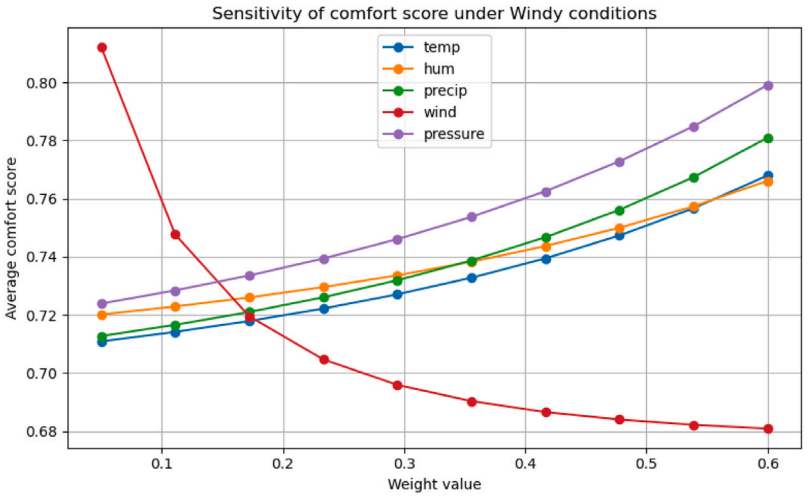


Fig. C.3. OAT sensitivity of the comfort score under Windy conditions. One weight is varied while the remaining weights are re-normalized to keep $\sum_i w_i = 1$; the score is averaged over a multi-point meteorological grid.

weight exhibits relatively mild sensitivity under sunny conditions. These observations indicate that the metric responds consistently to regime-dominant discomfort sources rather than being dominated by arbitrary parameter choices.

Data availability

Data will be made available on request.

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